vakint p-integrals: RustRed vs Forcer

clean benchmark `bench-clean-batch4-pushed` (`bench-clean.v1`) · 2026-11-13 · itphlies · document commit base gammaloop `da7e31d57`, rustred `6992e2dd` · measured build gammaloop `da7e31d57`, rustred `6992e2dd` (identical: last section)

In short. One harness, one build and one data set for every number: 630 rows (cell × backend × threads × state, medians of 1890 measurements in 90 runs, 18:31–06:28Z), 1L–4L, RustRed (M5 batch-4 package) against Forcer, at 1 and 16 threads, cold and warm, time and peak RSS. Older benchmark rows are not mixed in.

- **Time.** RustRed is slower than Forcer (cold evaluate) in 0 of 63 cells at 1 thread and in 1 at 16 threads (`physics_L4` ×1.16) in every other cell it is faster. From 1 to 16 threads the 4L cold evaluate changes by a median ×0.64 for RustRed and ×1.16 for Forcer (`tform -w16`).
- **16-thread numbers are upper bounds.** The host was shared and loaded: on the 16 CPUs of a 16-thread job unpinned single-thread processes of other users took a median of 0.480 cores per job (median over 43 jobs; up to 10.4 cores in single samples). Both backends were exposed alike; 1-thread jobs ran with their SMT sibling held idle; other users' load on that sibling: median < 0.001 cores.
- **Memory.** At 1L–3L RustRed's peak RSS is below Forcer's (process plus FORM) in 94 of 94 cells × thread counts (median ×0.73). At 4L it is above Forcer's in 15 of 16 cells (below: `physics_L4` ×0.64 / ×0.30): median ×2.97 at 1 thread and ×2.85 at 16 threads, worst `L4_d1_r8` (1.3 GiB against 206 MiB, ×6.68) and `L4_d1_r8` at 16 threads (1.9 GiB against 353 MiB, ×5.39).
- **Results.** All 606 rows with a pinned reference are byte-identical to it (the outputs pinned by benchmark M3 in `data/p_results.json`, re-checked when this document is built); 24 RustRed rows (`L4_d0_r6`, `L4_d0_r8`, `L4_d1_r8`, `L4_d2_r8`) have no pinned reference (M3 capped them); every cell gave one output per backend across states, thread counts and repetitions.

Table 1: RustRed over Forcer per loop count and thread count, medians over the cells (grid and physics cell) of the per-cell medians over repetitions. Cold evaluate: wall time of `Vakint::evaluate`. Cold process: the whole RustRed process (load, warm-up, evaluate, exit) without its cache save, over Forcer's whole process. Peak RSS: RustRed's process against Forcer's process plus its FORM subprocess. Slower: cold evaluate ratio > 1.

L	thr.	cells	cold evaluate		cold process		peak RSS		slower than Forcer
			median	worst	median	worst	median	worst	
1 L	1	15	×0.01	×0.02 (L1 d1 r8)	×0.18	×0.19	×0.72	×0.73 (L1 d1 r0)	none
1 L	16	15	×0.01	×0.07 (L1 d1 r8)	×0.13	×0.17	×0.72	×0.76 (L1 d2 r0)	none
2 L	1	16	×0.01	×0.05 (physics L2)	×0.18	×0.20	×0.73	×0.75 (L2 d1 r0)	none
2 L	16	16	×0.02	×0.11 (physics L2)	×0.18	×0.19	×0.74	×0.78 (L2 d2 r4)	none
3 L	1	16	×0.02	×0.30 (physics L3)	×0.08	×0.30	×0.73	×0.75 (L3 d2 r2)	none
3 L	16	16	×0.02	×0.40 (physics L3)	×0.07	×0.38	×0.72	×0.79 (L3 d2 r0)	none
4 L	1	16	×0.07	×0.54 (physics L4)	×0.10	×0.55	×2.97	×6.68 (L4 d1 r8)	none
4 L	16	16	×0.03	×1.16 (physics L4)	×0.05	×1.16	×2.85	×5.39 (L4 d1 r8)	**physics L4**



Reading guide (Tables 2 and 3). Each cell sums all tops of that loop count at the given dots, times a numerator of that rank. Bold: RustRed cold (seconds, median of 3). F: Forcer cold; ×: RustRed cold over Forcer cold (colour); ws: RustRed warm, a fresh process on its own cold cache; wp: the warm repeat inside the cold process; Fw: Forcer warm (materialised). RSS: RustRed's cold peak RSS and its ratio to Forcer's (red: RustRed higher). ✓: result exactly equal to Forcer's (M3 exact tier, `p_exact.json`, carried by the byte-identical outputs); n.c.: not exact-checked (cost). †: two repetitions instead of three.

Table 2: 1 thread (`n_cores` 1, `form`); RustRed arithmetic `modular_polynomial`.

L	dots	rank 0	rank 2	rank 4	rank 6	rank 8
1 L	0	**0.004** ✓ F 0.434 **×1/109** ws 0.004 wp 0.003 Fw 0.438 RSS 108 MiB ×0.71	**0.005** ✓ F 0.426 **×0.01** ws 0.004 wp 0.004 Fw 0.433 RSS 108 MiB ×0.72	**0.004** ✓ F 0.442 **×1/107** ws 0.004 wp 0.004 Fw 0.439 RSS 108 MiB ×0.71	**0.005** ✓ F 0.442 **×0.01** ws 0.005 wp 0.005 Fw 0.442 RSS 108 MiB ×0.71	**0.004** ✓ F 0.456 **×1/103** ws 0.005 wp 0.004 Fw 0.446 RSS 108 MiB ×0.72
	1	**0.004** ✓ F 0.440 **×1/107** ws 0.004 wp 0.004 Fw 0.433 RSS 108 MiB ×0.73	**0.006** ✓ F 0.444 **×0.01** ws 0.005 wp 0.005 Fw 0.446 RSS 108 MiB ×0.72	**0.007** ✓ F 0.448 **×0.02** ws 0.007 wp 0.007 Fw 0.441 RSS 108 MiB ×0.71	**0.009** ✓ F 0.438 **×0.02** ws 0.009 wp 0.008 Fw 0.448 RSS 108 MiB ×0.73	**0.011** ✓ F 0.453 **×0.02** ws 0.011 wp 0.010 Fw 0.458 RSS 108 MiB ×0.72
	2	**0.004** ✓ F 0.445 **×1/111** ws 0.004 wp 0.004 Fw 0.441 RSS 108 MiB ×0.73	**0.004** ✓ F 0.444 **×1/101** ws 0.004 wp 0.004 Fw 0.445 RSS 108 MiB ×0.72	**0.005** ✓ F 0.446 **×0.01** ws 0.004 wp 0.005 Fw 0.444 RSS 108 MiB ×0.71	**0.005** ✓ F 0.449 **×0.01** ws 0.005 wp 0.004 Fw 0.454 RSS 108 MiB ×0.71	**0.005** ✓ F 0.445 **×0.01** ws 0.005 wp 0.004 Fw 0.440 RSS 108 MiB ×0.72
2 L	0	**0.006** ✓ F 0.476 **×0.01** ws 0.006 wp 0.005 Fw 0.485 RSS 108 MiB ×0.73	**0.006** ✓ F 0.484 **×0.01** ws 0.005 wp 0.005 Fw 0.486 RSS 108 MiB ×0.73	**0.006** ✓ F 0.468 **×0.01** ws 0.005 wp 0.005 Fw 0.477 RSS 108 MiB ×0.71	**0.008** ✓ F 0.477 **×0.02** ws 0.006 wp 0.006 Fw 0.479 RSS 108 MiB ×0.72	**0.009** ✓ F 0.490 **×0.02** ws 0.008 wp 0.006 Fw 0.499 RSS 108 MiB ×0.73
	1	**0.006** ✓ F 0.483 **×0.01** ws 0.006 wp 0.005 Fw 0.471 RSS 108 MiB ×0.75	**0.007** ✓ F 0.489 **×0.01** ws 0.006 wp 0.005 Fw 0.484 RSS 108 MiB ×0.72	**0.007** ✓ F 0.474 **×0.01** ws 0.006 wp 0.005 Fw 0.491 RSS 108 MiB ×0.73	**0.008** ✓ F 0.470 **×0.02** ws 0.006 wp 0.006 Fw 0.483 RSS 108 MiB ×0.71	**0.010** ✓ F 0.505 **×0.02** ws 0.007 wp 0.006 Fw 0.497 RSS 108 MiB ×0.73
	2	**0.007** ✓ F 0.476 **×0.01** ws 0.006 wp 0.005 Fw 0.474 RSS 108 MiB ×0.72	**0.007** ✓ F 0.482 **×0.01** ws 0.006 wp 0.005 Fw 0.482 RSS 108 MiB ×0.73	**0.007** ✓ F 0.493 **×0.01** ws 0.007 wp 0.006 Fw 0.481 RSS 108 MiB ×0.71	**0.009** ✓ F 0.497 **×0.02** ws 0.007 wp 0.007 Fw 0.492 RSS 108 MiB ×0.73	**0.012** ✓ F 0.514 **×0.02** ws 0.009 wp 0.008 Fw 0.511 RSS 108 MiB ×0.71
3 L	0	**0.028** ✓ F 1.77 **×0.02** ws 0.026 wp 0.008 Fw 1.79 RSS 105 MiB ×0.70	**0.037** ✓ F 2.01 **×0.02** ws 0.029 wp 0.011 Fw 1.98 RSS 105 MiB ×0.71	**0.031** ✓ F 2.32 **×0.01** ws 0.021 wp 0.010 Fw 2.46 RSS 105 MiB ×0.69	**0.099** ✓ F 2.73 **×0.04** ws 0.035 wp 0.023 Fw 2.58 RSS 105 MiB ×0.69	**0.201** ✓ F 3.00 **×0.07** ws 0.053 wp 0.040 Fw 3.11 RSS 105 MiB ×0.70
	1	**0.029** ✓ F 2.06 **×0.01** ws 0.027 wp 0.010 Fw 2.07 RSS 108 MiB ×0.73	**0.039** ✓ F 2.12 **×0.02** ws 0.029 wp 0.012 Fw 2.07 RSS 108 MiB ×0.73	**0.043** ✓ F 2.26 **×0.02** ws 0.033 wp 0.013 Fw 2.26 RSS 108 MiB ×0.75	**0.124** ✓ F 2.80 **×0.04** ws 0.065 wp 0.039 Fw 2.66 RSS 108 MiB ×0.73	**0.296** ✓ F 3.14 **×0.09** ws 0.131 wp 0.111 Fw 3.34 RSS 108 MiB ×0.72
	2	**0.031** ✓ F 2.06 **×0.02** ws 0.028 wp 0.010 Fw 2.05 RSS 111 MiB ×0.74	**0.043** ✓ F 2.21 **×0.02** ws 0.035 wp 0.016 Fw 2.23 RSS 111 MiB ×0.75	**0.048** ✓ F 2.31 **×0.02** ws 0.036 wp 0.015 Fw 2.31 RSS 111 MiB ×0.75	**0.116** ✓ F 2.74 **×0.04** ws 0.065 wp 0.046 Fw 2.73 RSS 111 MiB ×0.75	**0.272** ✓ F 3.49 **×0.08** ws 0.126 wp 0.108 Fw 3.52 RSS 111 MiB ×0.73
4 L	0	**1.29** ✓ F 23.5 **×0.06** ws 1.09 wp 0.155 Fw 25.4 RSS 302 MiB ×2.06	**1.62** ✓ F 36.4 **×0.04** ws 1.24 wp 0.227 Fw 36.4 RSS 348 MiB ×2.37	**1.76** n.c. F 47.5 **×0.04** ws 1.17 wp 0.235 Fw 47.3 RSS 415 MiB ×2.88	**10.4** n.c. F 104 **×0.10** ws 4.02 wp 3.03 Fw 107 RSS 861 MiB ×4.55	**25.3** n.c. F 136 **×0.19** ws 10.2 wp 9.39 Fw 136 RSS 1.1 GiB ×5.15
	1	**1.72** ✓ F 35.8 **×0.05** ws 1.40 wp 0.429 Fw 36.1 RSS 336 MiB ×2.43	**2.52** ✓ F 52.2 **×0.05** ws 1.81 wp 0.781 Fw 50.4 RSS 381 MiB ×2.49	**2.00** n.c. F 50.1 **×0.04** ws 1.37 wp 0.394 Fw 50.6 RSS 429 MiB ×3.04	**12.9** n.c. F 100 **×0.13** ws 6.17 wp 5.51 Fw 99.6 RSS 912 MiB ×4.82	**35.8** n.c. F 142 **×0.25** ws 19.4 wp 22.5 Fw 144 RSS 1.3 GiB ×6.68
	2	**4.05** ✓ F 46.1 **×0.09** ws 3.22 wp 1.50 Fw 46.1 RSS 375 MiB ×2.28	**2.80** ✓ F 48.9 **×0.06** ws 2.02 wp 0.947 Fw 48.5 RSS 410 MiB ×2.90	**2.40** n.c. F 59.6 **×0.04** ws 1.61 wp 0.603 Fw 59.6 RSS 456 MiB ×3.23	**18.1** n.c. F 124 **×0.15** ws 10.1 wp 9.15 Fw 125 RSS 1.0 GiB ×5.06	**58.9** n.c. F 197 **×0.30** ws 34.1 wp 27.2 Fw 195 RSS 1.6 GiB ×5.89
phys.		**2 L 0.024** ✓ F 0.490 **×0.05** ws 0.021 wp 0.020 Fw 0.504 RSS 108 MiB ×0.73	**3 L 0.416** ✓ F 1.39 **×0.30** ws 0.399 wp 0.385 Fw 1.39 RSS 108 MiB ×0.71	**4 L 69.6** ✓ F 128 **×0.54** ws 69.3 wp 68.7 Fw 127 RSS 443 MiB ×0.64	5L: not part of this benchmark	

Table 3: 16 threads (`n_cores` 16, `tform -w16`) on 16 CPUs of one NUMA node; upper bounds (foreign load, Table 1 note); RustRed arithmetic `modular_polynomial`.

L	dots	rank 0	rank 2	rank 4	rank 6	rank 8
1 L	0	**0.007** ✓ F 0.672 **×1/100** ws 0.006 wp 0.007 Fw 0.526 RSS 102 MiB ×0.71	**0.008** ✓ F 0.543 **×0.02** ws 0.006 wp 0.007 Fw 0.459 RSS 102 MiB ×0.74	**0.007** ✓ F 0.499 **×0.01** ws 0.006 wp 0.006 Fw 0.650 RSS 102 MiB ×0.72	**0.007** ✓ F 0.441 **×0.02** ws 0.007 wp 0.007 Fw 0.533 RSS 102 MiB ×0.72	**0.007** ✓ F 0.527 **×0.01** ws 0.008 wp 0.008 Fw 0.650 RSS 102 MiB ×0.72
	1	**0.007** ✓ F 0.524 **×0.01** ws 0.006 wp 0.006 Fw 0.535 RSS 102 MiB ×0.72	**0.012** ✓ F 0.552 **×0.02** ws 0.011 wp 0.012 Fw 0.655 RSS 102 MiB ×0.71	**0.018** ✓ F 0.513 **×0.04** ws 0.017 wp 0.018 Fw 0.445 RSS 102 MiB ×0.72	**0.024** ✓ F 0.495 **×0.05** ws 0.026 wp 0.022 Fw 0.499 RSS 102 MiB ×0.74	**0.034** ✓ F 0.463 **×0.07** ws 0.031 wp 0.032 Fw 0.591 RSS 102 MiB ×0.72
	2	**0.006** ✓ F 0.488 **×0.01** ws 0.006 wp 0.007 Fw 0.565 RSS 102 MiB ×0.76	**0.007** ✓ F 0.742 **×1/107** ws 0.006 wp 0.006 Fw 0.702 RSS 102 MiB ×0.75	**0.007** ✓ F 0.668 **×0.01** ws 0.007 wp 0.007 Fw 0.518 RSS 102 MiB ×0.72	**0.007** ✓ F 0.553 **×0.01** ws 0.007 wp 0.006 Fw 0.505 RSS 102 MiB ×0.71	**0.010** ✓ F 0.579 **×0.02** ws 0.007 wp 0.008 Fw 0.646 RSS 102 MiB ×0.71
2 L	0	**0.008** ✓ F 0.488 **×0.02** ws 0.010 wp 0.007 Fw 0.501 RSS 105 MiB ×0.76	**0.009** ✓ F 0.481 **×0.02** ws 0.010 wp 0.008 Fw 0.475 RSS 105 MiB ×0.73	**0.011** ✓ F 0.480 **×0.02** ws 0.008 wp 0.009 Fw 0.496 RSS 105 MiB ×0.76	**0.014** ✓ F 0.486 **×0.03** ws 0.009 wp 0.011 Fw 0.508 RSS 105 MiB ×0.74	**0.015** ✓ F 0.515 **×0.03** ws 0.012 wp 0.012 Fw 0.502 RSS 105 MiB ×0.74
	1	**0.009** ✓ F 0.486 **×0.02** ws 0.009 wp 0.009 Fw 0.511 RSS 105 MiB ×0.73	**0.010** ✓ F 0.499 **×0.02** ws 0.009 wp 0.008 Fw 0.501 RSS 105 MiB ×0.74	**0.009** ✓ F 0.507 **×0.02** ws 0.009 wp 0.008 Fw 0.505 RSS 105 MiB ×0.74	**0.012** ✓ F 0.542 **×0.02** ws 0.012 wp 0.011 Fw 0.534 RSS 105 MiB ×0.73	**0.014** ✓ F 0.520 **×0.03** ws 0.012 wp 0.010 Fw 0.506 RSS 105 MiB ×0.74
	2	**0.010** ✓ F 0.497 **×0.02** ws 0.009 wp 0.009 Fw 0.497 RSS 105 MiB ×0.74	**0.012** ✓ F 0.509 **×0.02** ws 0.011 wp 0.010 Fw 0.504 RSS 105 MiB ×0.71	**0.012** ✓ F 0.496 **×0.02** ws 0.012 wp 0.009 Fw 0.511 RSS 105 MiB ×0.78	**0.012** ✓ F 0.503 **×0.02** ws 0.012 wp 0.009 Fw 0.511 RSS 105 MiB ×0.74	**0.015** ✓ F 0.521 **×0.03** ws 0.014 wp 0.012 Fw 0.547 RSS 105 MiB ×0.73
3 L	0	**0.029** ✓ F 2.22 **×0.01** ws 0.025 wp 0.014 Fw 2.29 RSS 96 MiB ×0.67	**0.034** ✓ F 2.33 **×0.01** ws 0.029 wp 0.016 Fw 2.06 RSS 96 MiB ×0.68	**0.025** ✓ F 2.81 **×1/112** ws 0.019 wp 0.017 Fw 2.94 RSS 96 MiB ×0.68	**0.059** ✓ F 3.20 **×0.02** ws 0.038 wp 0.033 Fw 3.30 RSS 96 MiB ×0.64	**0.097** ✓ F 3.39 **×0.03** ws 0.069 wp 0.050 Fw 3.12 RSS 96 MiB ×0.62
	1	**0.030** ✓ F 2.45 **×0.01** ws 0.030 wp 0.014 Fw 3.15 RSS 105 MiB ×0.76	**0.037** ✓ F 2.50 **×0.01** ws 0.035 wp 0.021 Fw 2.17 RSS 105 MiB ×0.74	**0.048** ✓ F 2.79 **×0.02** ws 0.036 wp 0.022 Fw 2.57 RSS 105 MiB ×0.73	**0.098** ✓ F 3.02 **×0.03** ws 0.070 wp 0.057 Fw 3.16 RSS 105 MiB ×0.71	**0.213** ✓ F 3.87 **×0.06** ws 0.130 wp 0.108 Fw 3.74 RSS 105 MiB ×0.69
	2	**0.038** ✓ F 2.77 **×0.01** ws 0.040 wp 0.022 Fw 2.97 RSS 105 MiB ×0.79	**0.059** ✓ F 2.83 **×0.02** ws 0.052 wp 0.030 Fw 2.57 RSS 105 MiB ×0.76	**0.055** ✓ F 3.15 **×0.02** ws 0.053 wp 0.030 Fw 3.51 RSS 105 MiB ×0.75	**0.150** ✓ F 3.33 **×0.05** ws 0.107 wp 0.081 Fw 3.86 RSS 105 MiB ×0.74	**0.280** ✓ F 4.77 **×0.06** ws 0.217 wp 0.191 Fw 4.28 RSS 105 MiB ×0.73
4 L	0	**0.841** ✓ F 28.9 **×0.03** ws 0.865 wp 0.177 Fw 32.0 RSS 344 MiB ×2.34	**1.01** ✓ F 50.0 **×0.02** ws 0.966 wp 0.241 Fw 51.4 RSS 402 MiB ×2.31	**1.14** n.c. F 63.1 **×0.02** ws 1.04 wp 0.276 Fw 61.1 RSS 468 MiB ×2.74	**4.49** n.c. F 95.4 **×0.05** ws 2.84 wp 1.66 Fw 104 RSS 1.1 GiB ×3.95	**7.72** n.c. F 137 **×0.06** ws 4.69 wp 3.42 Fw 136 RSS 1.5 GiB ×4.32
	1	**1.21** ✓ F 48.7 **×0.02** ws 1.18 wp 0.409 Fw 51.5 RSS 429 MiB ×2.52	**1.82** ✓ F 62.7 **×0.03** ws 1.67 wp 0.638 Fw 52.3 RSS 491 MiB ×2.52	**1.87** n.c. F 78.7 **×0.02** ws 1.45 wp 0.487 Fw 72.1 RSS 496 MiB ×2.85	**5.34** n.c. F 112 **×0.05** ws 3.70 wp 2.45 Fw 113 RSS 1.2 GiB ×4.26	**12.4** n.c. F 141 **×0.09** ws 8.14 wp 7.75 Fw 154 RSS 1.9 GiB ×5.39
	2	**1.77** ✓ F 58.3 **×0.03** ws 1.53 wp 0.717 Fw 54.8 RSS 505 MiB ×2.12	**2.03** ✓ F 64.1 **×0.03** ws 1.59 wp 0.782 Fw 59.9 RSS 540 MiB ×2.86	**1.74** n.c. F 63.6 **×0.03** ws 1.59 wp 0.794 Fw 63.4 RSS 570 MiB ×3.11	**8.45** n.c. F 135 **×0.06** ws 5.93 wp 4.72 Fw 132 RSS 1.4 GiB ×4.75	**18.5** n.c. F 176 **×0.11** ws 12.1 wp 9.81 Fw 178 RSS 2.2 GiB ×4.95
phys.		2 L **0.056** ✓ F 0.528 **×0.11** ws 0.051 wp 0.051 Fw 0.545 RSS 105 MiB ×0.74	3 L **0.708** ✓ F 1.78 **×0.40** ws 0.641 wp 0.752 Fw 1.89 RSS 99 MiB ×0.67	4 L **101** ✓ F 87.5 **×1.16** ws 105 wp 96.1 Fw 75.1 RSS 382 MiB ×0.30	5L: not part of this benchmark	

Table 4: 4L in detail, cold runs, medians. RustRed: evaluate (the timed call), its reduction span, the load of engine and library before the clock, the cache save after the clock, and the whole process without that save. Forcer: evaluate and its whole process. Foreign cores: median load of other users' processes on the job's CPUs (16 threads only; 1-thread jobs had none to speak of).

cell	thr.	RustRed cold (s)					Forcer cold (s)		RustRed / Forcer		peak RSS			foreign cores RR / F
		evaluate	reduc-tion	load	save	process − save	evaluate	process	evaluate	process	RustRed	Forcer	ratio	
L4 d0 r0	1	**1.29**	1.26	0.653	0.063	2.02	**23.5**	24.5	**×0.06**	×0.08	302 MiB	147 MiB	×2.06	—
L4 d0 r0	16	**0.841**	0.800	0.844	0.088	1.75	**28.9**	29.6	**×0.03**	×0.06	344 MiB	147 MiB	×2.34	0.3 / 1.2
L4 d0 r2	1	**1.62**	1.58	0.720	0.088	2.40	**36.4**	37.1	**×0.04**	×0.06	348 MiB	147 MiB	×2.37	—
L4 d0 r2	16	**1.01**	0.968	0.812	0.118	1.99	**50.0**	51.0	**×0.02**	×0.04	402 MiB	174 MiB	×2.31	1.2 / 1.7
L4 d0 r4	1	**1.76**	1.72	0.686	0.112	2.60	**47.5**	48.1	**×0.04**	×0.05	415 MiB	144 MiB	×2.88	—
L4 d0 r4	16	**1.14**	1.09	0.768	0.147	2.13	**63.1**	63.9	**×0.02**	×0.03	468 MiB	171 MiB	×2.74	0.1 / 0.1
L4 d0 r6	1	**10.4**	10.4	0.713	1.05	11.4	**104**	105	**×0.10**	×0.11	861 MiB	189 MiB	×4.55	—
L4 d0 r6	16	**4.49**	4.39	1.03	1.00	5.67	**95.4**	96.0	**×0.05**	×0.06	1.1 GiB	291 MiB	×3.95	0.4 / 0.3
L4 d0 r8	1	**25.3**	25.2	0.762	1.69	26.5	**136**	136	**×0.19**	×0.19	1.1 GiB	224 MiB	×5.15	—
L4 d0 r8	16	**7.72**	7.60	0.992	1.97	9.09	**137**	138	**×0.06**	×0.07	1.5 GiB	367 MiB	×4.32	1.1 / 1.1
L4 d1 r0	1	**1.72**	1.68	0.705	0.136	2.51	**35.8**	36.4	**×0.05**	×0.07	336 MiB	138 MiB	×2.43	—
L4 d1 r0	16	**1.21**	1.16	0.909	0.190	2.21	**48.7**	49.6	**×0.02**	×0.04	429 MiB	170 MiB	×2.52	0.4 / 0.2
L4 d1 r2	1	**2.52**	2.46	0.748	0.198	3.34	**52.2**	52.9	**×0.05**	×0.06	381 MiB	153 MiB	×2.49	—
L4 d1 r2	16	**1.82**	1.75	0.732	0.182	2.63	**62.7**	63.4	**×0.03**	×0.04	491 MiB	195 MiB	×2.52	0.1 / 2.3
L4 d1 r4	1	**2.00**	1.94	0.674	0.137	2.74	**50.1**	50.7	**×0.04**	×0.05	429 MiB	141 MiB	×3.04	—
L4 d1 r4	16	**1.87**	1.77	1.05	0.228	3.03	**78.7**	79.6	**×0.02**	×0.04	496 MiB	174 MiB	×2.85	1.1 / 1.0
L4 d1 r6	1	**12.9**	12.7	0.764	1.17	13.9	**100**	101	**×0.13**	×0.14	912 MiB	189 MiB	×4.82	—
L4 d1 r6	16	**5.34**	5.03	0.784	1.35	6.48	**112**	113	**×0.05**	×0.06	1.2 GiB	288 MiB	×4.26	0.3 / 0.3
L4 d1 r8	1	**35.8**	33.4	1.06	2.55	37.4	**142**	143	**×0.25**	×0.26	1.3 GiB	206 MiB	×6.68	—
L4 d1 r8	16	**12.4**	10.8	0.903	2.32	13.6	**141**	141	**×0.09**	×0.10	1.9 GiB	353 MiB	×5.39	1.8 / 3.7
L4 d2 r0	1	**4.05**	3.94	0.915	0.390	5.08	**46.1**	46.7	**×0.09**	×0.11	375 MiB	165 MiB	×2.28	—
L4 d2 r0	16	**1.77**	1.68	0.909	0.319	2.75	**58.3**	58.9	**×0.03**	×0.05	505 MiB	239 MiB	×2.12	0.3 / 1.2
L4 d2 r2	1	**2.80**	2.74	0.738	0.256	3.63	**48.9**	49.4	**×0.06**	×0.07	410 MiB	141 MiB	×2.90	—
L4 d2 r2	16	**2.03**	1.94	0.766	0.399	2.95	**64.1**	64.9	**×0.03**	×0.05	540 MiB	189 MiB	×2.86	1.2 / 0.2
L4 d2 r4	1	**2.40**	2.35	0.632	0.203	3.20	**59.6**	60.2	**×0.04**	×0.05	456 MiB	141 MiB	×3.23	—
L4 d2 r4	16	**1.74**	1.65	0.760	0.297	2.64	**63.6**	64.2	**×0.03**	×0.04	570 MiB	183 MiB	×3.11	0.0 / 0.7
L4 d2 r6	1	**18.1**	18.0	0.726	1.48	19.1	**124**	125	**×0.15**	×0.15	1.0 GiB	201 MiB	×5.06	—
L4 d2 r6	16	**8.45**	8.27	0.844	2.17	9.67	**135**	135	**×0.06**	×0.07	1.4 GiB	309 MiB	×4.75	0.8 / 0.3
L4 d2 r8	1	**58.9**	58.6	0.762	3.66	60.1	**197**	198	**×0.30**	×0.30	1.6 GiB	273 MiB	×5.89	—
L4 d2 r8	16	**18.5**	18.2	0.914	3.72	20.0	**176**	177	**×0.11**	×0.11	2.2 GiB	465 MiB	×4.95	0.1 / 0.5
physics L4	1	**69.6**	12.4	0.732	0.162	70.6	**128**	129	**×0.54**	×0.55	443 MiB	697 MiB	×0.64	—
physics L4	16	**101**	18.1	0.839	0.261	103	**87.5**	88.3	**×1.16**	×1.16	382 MiB	1.2 GiB	×0.30	4.9 / 1.4

Measured build and pushed tip

The rows were measured with the pushed tip itself: one frozen harness binary (sha256 `89d46d0e46b9382f`…) built with `just vakint-p-benchmark-build` from gammaloop `da7e31d57` (stamp = that commit, clean tree) pinning rustred `6992e2dd` the 4L library is `p_integral/p-library-1-4` of vakint-external-data `d9a55ff8` (p_integral-m5-batch4 d9a55ff8, v0.2.3 + batch-4 p area) read through `--external-data` (`42f9132a692fbfa7`…, p_integral/p-library-1-4 of vakint-external-data d9a55ff8: 474 files byte-identical (diff -rq) to artefacts/m5/batch4/p-library-1-4, the package the final binary regenerates in assemble4 check mode), the 1L–3L library the built-in one (identity ends `P_ORDERING_VERSION=1:generation`). No equivalence argument is needed. Before the hand-over (notes m5-batch4): test_vakint 1097/1097, R* 60/60, T-FORCER 22/22 tops of L4 d0_r8/d1_r8 through vakint, assemble4 check 4/0 and lint on the final tips; these verify results, not timings. gammaloop p_integral-vk-physics is not in batch 4.

Method. evaluate = Vakint::evaluate wall after the load and one warm-up bubble; the RustRed cache save (fsync) runs after the clock (save_seconds), then the in-process warm repeat; process = whole child process minus that repeat (load + warm-up + evaluate + save + exit); process_excl_save = process - save. Work directory /dev/shm (tmpfs) whole harness under taskset (1 CPU at 1 thread, its SMT sibling held idle by a flock; 16 CPUs of one NUMA node at 16 threads) and numactl --membind=<node>; 1-thread jobs in parallel on separate cores, 16-thread jobs one (late: two, on different NUMA nodes) at a time. States: RustRed cold (empty ordered-cache directory, shipped caches off), warm-same (a fresh process on a read-only copy of its cold cache), warm-process (in the cold process, after the save); Forcer cold (private empty materialisation directory) and warm (materialised); Forcer has no result cache. Three repetitions per state (†: two), median; cap 3600 s per child (no row was capped or skipped). OS pages are not flushed. Physics row: massless-quark (L−1)-rung gluon self-energy, Feynman gauge, transverse projector, external momentum kept.

Provenance. `data/p_clean.json` = artefact `bench-clean-batch4-pushed`, written `2026-11-13T06:29:15Z`, copied verbatim; host itphlies (AMD EPYC 9754 128-Core Processor), `FORM 5.0.0`, Forcer `68a30bf`; RustRed at 1L–3L with the built-in `p-library-1-3`, at 4L with `p-library-1-4` listed by external data. Outputs compared with the output_check: pinned M3 outputs; previous_build_check: every output of the combined bench-clean v1 data set, its appended batch-2 build and the pushed batch-2 data set (bench_clean.batch2-pushed.v1.json). M3: gammaloop fd8b44641 crates/vakint/docs/benchmark/data/p_results.json (output_sha256 per cell and backend). Exact tier: `data/p_exact.json` (gammaloop `1a8862a97`, rustred `b7fd9f2b`, 54 cells agree); of the 4L cells it marks n.c., the gate cross-check above covers the 22 tops of L4 d0_r8/d1_r8. The M3 benchmark rows remain in `data/p_results.json` for their pinned outputs only; none of their timings is used here.